

Spatial and size distributions of early recruits of *Cucumaria frondosa* in a coastal nursery (Nunavut, Canadian Arctic)

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ABSTRACT

The discovery of nursery sites of the sea cucumber *Cucumaria frondosa* in the Canadian Arctic offers opportunities to gather knowledge on the ecology of juveniles. Here, their size and spatial distributions were examined and inter-individual and inter-specific relations were assessed. Out of 893 juveniles collected between 1 and 3 m depths in the fall, 95% were early recruits (<1-year-old). They chiefly occurred on rubbles (~80%) representing the main substratum (~44%) in the nursery. The length of early recruits across all depths was ~1520 µm, with the smallest found at 3 m (~1407 µm). Fewer recruits were identified in the following spring (51.5%), allowing the calculation of a growth rate (~114 µm mo⁻¹). Their density was consistently ~7400 ind m² across depths. Juveniles were mostly found aggregated on flat surfaces (62.7%) or in shallow depressions (18.4%) on the underside of the stones, but also occurred on seaweed holdfasts (2.7%). The average nearest neighbour distance was ~0.37 cm, with some juveniles touching each other. More individuals were observed with their body oriented in the same direction (45%) than in opposite directions (25%). Four sympatric species were identified on the same substratum; the polychaete *Spirorbis spirorbis* (49.7%), encrusting bryozoans in order Cheilostomatida (40.5%), the limpet *Testudinalia testudinalis* (6.9%), and calcareous encrusting algae (2.9%). Data on newly settled sea cucumbers in the wild are rare. The present findings provide essential information on the population dynamics and carrying capacity of *C. frondosa* in support of local subsistence harvests and emerging small-scale commercial fisheries led by the Inuit community.

1. Introduction

Most sea cucumbers reproduce by broadcast spawning, releasing oocytes and spermatozoa in the water column (Mercier and Hamel, 2009). The eggs undergo either planktotrophic or lecithotrophic development (McEdward and Miner, 2001; Raff and Byrne, 2006; Sewell, Hamel and Mercier, 2024). In both cases, the pentactula settles after transitioning from a pelagic to a benthic lifestyle (e.g. Hamel and Mercier, 1996a). Knowledge of the distribution and ecology of early recruits (immediately post-settlement) of sea cucumbers in the wild environment is limited, mainly due to their tiny size and cryptic nature. On the few occasions that early recruits were observed in the wild, they rarely occurred on the same substratum and/or habitat as adults (Mercier

et al., 2000; Shiell, 2004; Yamana, Hamano and Miki, 2006; Wolfe, Mercier and Hamel, 2024). For instance, Mercier et al. (2000) observed early recruits (as small as ~1.7 mm long) of *Holothuria scabra* in the Solomon Islands settled on seagrass leaves before migrating to the surrounding sand (when reaching a length of 9–10 mm) where larger size individuals were also found. A unique reproductive strategy was discovered in Belize, whereby adults of *Holothuria floridana* were observed to foster offspring development on their body wall, from eggs to early juveniles, with some developing on the nearby seagrass leaves (Rogers et al., 2021). In most other studies, the so-called juveniles refer to substantially larger individuals and therefore would not be considered as early recruits (Wolfe et al., 2024).

Cucumaria frondosa is a cold-water sea cucumber with a distribution

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range extending from the North Atlantic to the Arctic ocean (Mercier et al., 2023). Individuals spawn once a year with external fertilisation leading to the development of pelagic lecithotrophic larvae which settle after ~45 days in the plankton (Hamel and Mercier, 1995, 1996a, 1996b). This species is locally harvested by northern communities in the Canadian Arctic as a subsistence food item (Wein, Freeman and Makus, 1996), specifically by the Inuit of Qikiqtait (traditional name of the Belcher Islands), an archipelago composed of ~1500 islands located in southeastern Hudson Bay. Small-scale commercial sea cucumber fisheries initiatives led by the Inuit are being explored as a new source of livelihood for the community (Hamel and Mercier, 2024). Meanwhile, concerns about the Arctic, including but not limited to, ice melting, raising temperature, degradation of the habitat, changes in species geographic distribution (Klein et al., 2008; Borgerson, 2013; Hanna et al., 2021; Florko et al., 2021) have led to the multiplication of marine protected areas across this region (Barry, Helgason and Guðmundsdóttir, 2017; Barry et al., 2023). The case of Qikiqtait is unique as the objectives there are multi-layered: the development of small-scale commercial fisheries while preserving sea cucumber stock for local consumption, and the creation of a marine protected area. Accordingly, some studies have emerged to support the community in their marine development and management, by co-generating reference data and guidelines.

The density of *C. frondosa* has been assessed using portable remotely operated vehicles (ROVs) in several locations around Qikiqtait, revealing variable densities: 3.1 and 3.4 ind m⁻² in the channels of Kataaluk (sea cucumber mean length 111 mm) and Katak (mean length 77 mm), respectively, 24.8 ind m⁻² (mean length 60 mm) along the shore of Sanikiluaq (Ma et al., 2025a) and up to 50 ind m⁻² in the Kataapik channel (Hamel and Mercier, 2024). The density of *C. frondosa* decreased to 1.3 ind m⁻² (mean length 170 mm) in the outer area of Kataaluk, 0.4 ind m⁻² (mean length 110 mm) in Puttaalalik and 0.1 ind m⁻² (mean length 100 mm) near Ivittuuq Island (Morrison, Hamel and Mercier, 2025). The average size of the sea cucumbers in these studies ranged between 60 and 170 mm, representing sub-adults or adults (Leyte et al., in prep.). These results may help to determine the most suitable location, i.e. with the greatest density of marketable sea cucumbers, to start a small-scale commercial fishery. Meanwhile, the nutritional composition of different organs and tissues of *C. frondosa* have been assessed in Ivittuuq, Kataapik and Kataaluk to identify which location to conserve for subsistence (sea cucumbers consumed fresh) versus commercial harvests (dry products), while respecting and preserving ancestral and traditional uses (Pierrat et al., 2025a). Moreover, individuals of *C. frondosa* sampled in Qikiqtait had a more interesting nutritional composition for human consumption than conspecifics from Atlantic Canada (Morrison et al., 2024), which seems to be promising for future commercialisation.

Recently, Hamel et al. (2023) identified and characterised the first ever discovered nursery ground of *C. frondosa* in the Arctic. The juveniles ranged from 0.9 and 20 mm, with the smallest ones considered early recruits that were less than one-year-old. Their density reached up to 104 ind m⁻², leading to the assumption that some 300 billion juveniles could be found in the entire archipelago of Qikiqtait at any time of the year. Early recruits were mostly found attached on live or dead blue mussels and the underside of stones (Hamel et al., 2023). Other settlement substrata were later defined in Qikiqtait, namely blades and holdfasts of seaweeds, coralline algae, encrusting bryozoans, bivalves and even barnacles (Ma et al., 2025b). The fact that larger individuals were found in deeper waters (Morrison et al., 2025; Ma et al., 2025a), whereas early recruits and juveniles were found in shallow waters, i.e. between 1 and 3 m depths, supports ontogenic migration (Hamel et al., 2023; Ma et al., 2025a) likely effected by actively adjusting buoyancy to move around with the currents (Hamel et al., 2019). Survival rates from the juvenile to the adult stage are unknown in the wild where predatory and environmental pressures on early juveniles are poorly documented. Recently, Pierrat et al. (2025b) identified a halacarid mite as a potential

predator of early juveniles of *C. frondosa* in Qikiqtait. Moreover, during laboratory experiments, So, Hamel and Mercier (2010) demonstrated that juveniles of the sea star *Solaster endeca* (also recorded in Qikiqtait waters; J-F.H. pers. comm.) could prey on early recruits of *C. frondosa* and can presumably reach the underside of substrata inaccessible to larger predators.

Identifying nursery habitats and obtaining data on the abundance and distribution of early recruits are essential steps in the fundamental understanding of adult populations, and in preventing stock depletion, especially in the context of new fishery initiatives. The main objective of the present study was to develop a greater understanding of the size-frequency distribution and spatial ecology of early recruits of *C. frondosa* in a wild population inside the nursery grounds (Qikiqtait, Nunavut, Canada). Vertical distribution of early recruits at depths ranging from 1 to 3 m was analysed, as well as their occurrence and precise location on different types of substrata. Inter-individual relations (dispersion, distance and orientation) were investigated. Finally, the presence of sympatric species on the same substrata was assessed. This study aims to support the creation and delimitation of a marine protected area in Qikiqtait while simultaneously informing small-scale fishery initiatives and preserving food security for the local community. More broadly, the findings also offer novel insights into the early life history of wild sea cucumbers, which is globally understudied (Wolfe et al., 2024) at a time when commercial species are facing increasing threats (Mercier et al., 2025).

2. Material & methods

2.1. Study site

The study was conducted in Qikiqtait, located in the southeast of the Hudson Bay (Nunavut, Canada), in the lower Canadian Arctic (Fig. 1). More precisely, sampling was carried out inside a nursery habitat of *C. frondosa* at depths ranging from 1 to 3 m in the shallow subtidal zone outside of the channel of Kataaluk (~530 m wide, located between Renouf and Wiegand Islands; 56°36'13.1"N; 79°08'30.4"W; Fig. 1). The main study occurred in October 2024 in the absence of sea ice cover, with more limited under-ice sampling added opportunistically in May 2025.

2.2. Habitat description

To characterise the general habitat in which juvenile sea cucumbers were found (type of substrata), three video surveys were conducted using a GoPro 5 camera. Sixteen distinct still frames were extracted from the videos and the percentage of surface covered by fine substrata (<0.5 cm), gravel (>0.5–8 cm), rubble (>8–30 cm), small boulders (>30–60 cm), large boulders (>60 cm) and seaweeds was assessed using Image J (version 1.54 g), based primarily on the substratum categories defined by Platts, Megahan and Minshall (1983). Sea water parameters (temperature, salinity and pH) were recorded at ~0.5 m depth using a multiprobe (YSI PRO 1030) at the beginning, middle and end of the sampling during October 2024.

2.3. Sampling and processing

In October 2024, stones harbouring juveniles (including early recruits) were collected in Kataaluk from a 7 m long boat. Local collaborators used a long pole net (traditional fishing tool), constituted of a wooden pole ~4 m long terminated by a round net (~35 cm diameter, mesh size of ~1.5 cm). A preliminary sample of stones was collected to confirm the presence of juveniles at the site, followed by 18 formal samplings, each one raking ~0.2 m² of the bottom, every 0.5 m between 1 and 3 m depths. Each sample contained a variable number of stones whose maximum length and width were recorded with a measuring tape. Then, stones were closely examined to find juvenile sea

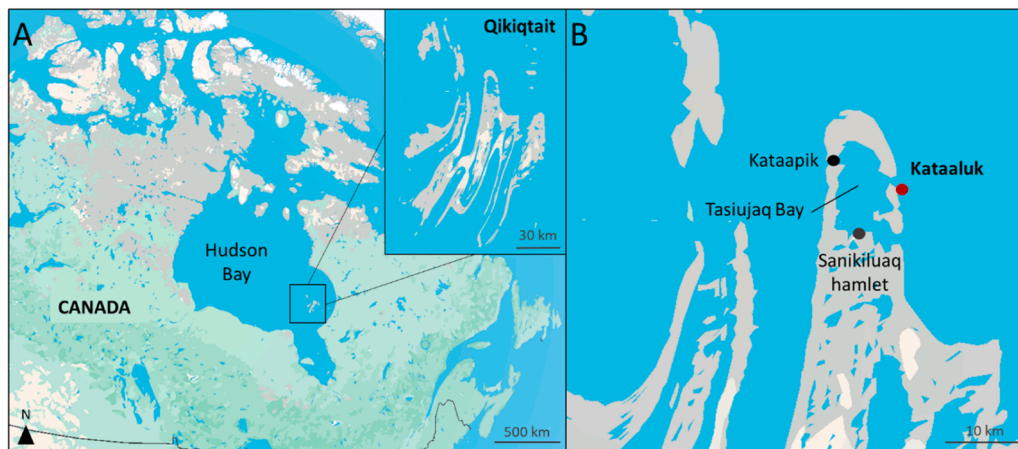


Fig. 1. (A) Location of Qikiqtait in the Hudson Bay (territory of Nunavut, northern Canada). (B) Location of the studied nursery of Kataaluk, the previously studied site of Kataapik and Tasiujaq Bay in the north of Qikiqtait, near the hamlet of Sanikiluaq.

cucumbers. When juveniles were present, the stones were individually photographed (general view and close-up, alongside a scale bar) with an Olympus TG6. The colonised surface area (surface occupied by the juveniles) was measured by drawing two perpendicular vectors with a ruler. Juveniles were counted, removed from the stones and stored in insulated containers during transport to the laboratory at the Qikiqtait Center (Sanikiluaq), where they were preserved in 70% ethanol for transport to the Department of Ocean Sciences (Memorial University, Newfoundland). There, the contracted length (from mouth to anus) of each juvenile was measured under a motorised stereomicroscope (Leica M205) using Leica Application Suite (LASX). According to the study by Hamel et al. (2023) in the same geographic area, juveniles were classified by age groups relative to their size, with a length $\leq 2000 \mu\text{m}$ representing early recruits in age class 0-I (<1 year old). Age class II included juveniles with lengths ranging from > 2000 – $6000 \mu\text{m}$, age class III were those with lengths from > 6000 – $10,000 \mu\text{m}$ and finally, juveniles $> 10,000 \mu\text{m}$ were considered as age class IV. This classification, where applicable, was also used to link the sea cucumber length with age throughout the present study. The samples collected in October 2024 were used to conduct a suite of analyses (see below).

In May 2025, another sampling was conducted at the same location in Kataaluk, but reached with a snowmobile. Since the site was still covered in thick sea ice (thickness ~ 75 cm), an auger was used to open six connected holes (to have a surface large enough to use the long pole net) where the water depth was ~ 2 m. A total of 18 samplings were made through the ice using the long pole net to collect stones. The May 2025 samples were examined as described earlier and used only to compare the size-frequency distributions of juveniles between seasons and to estimate growth rate.

2.4. Micro-habitat occupancy

To characterise the precise locations of juveniles on the substratum, 47 photos were analysed where both the substratum and sea cucumbers could be clearly seen. Five types of micro-habitats on stones were defined: (1) flat surfaces, (2) edges, (3) crevices, defined as narrow cracks (anfractuosités), deeper than wide or as deep as wide, (4) shallow depressions defined as broad, gently concave areas, and (5) seaweed holdfasts. For each photo, the number of juveniles in each location was recorded.

2.5. Inter-individual relationships

Index of dispersion. The dispersion of the juveniles was investigated using the variance-to-mean ratio method developed by Krebs (1999). This work was based on all the photos (10) where the ruler was correctly

aligned to the stone and was on the same plane, which represented ten different stones hosting 258 of the 847 juveniles (30% of the individuals). The method involved superimposing a 10×10 cm grid over each photo (a grid size determined during preliminary work) and counting the number of individuals in each square (Supplementary Figure S1). Globally, the total number of squares analysed on each photo was 100, which was twice more than the minimum square number recommended by Green (1966). An index of dispersion (I) was calculated as follow:

$$I = \frac{S^2}{\bar{x}}$$

where S^2 is the variance among grid squares and $\bar{x}$ the mean number of individuals per square. Values of $I > 1$ meant that individuals were aggregated, $I = 1$ meant that individuals were randomly dispersed, and $I < 1$ that individuals were uniformly dispersed. The main limit of the variance-to-mean ratio method is that some non-random patterns can also give $I = 1$ (Krebs, 1999). In this case, visual checks were made to ensure that the index was mainly due to the dispersion of individuals rather than a misleading combination of variances.

Distance. The distance between juveniles on the same stone was measured on the 17 photos where the ruler was correctly aligned to the stone and was on the same plane. Using Image J software, vectors of the closest distance between 350 pairs of juveniles were drawn. The length (from mouth to anus) of each juvenile was measured as well.

Orientation. The orientation of the juveniles relative to each other (position of the crown of tentacles or anterior section of the body when tentacles were not deployed) was analysed using the 21 close-up photos of different groups of sea cucumbers on which their anterior and posterior ends were distinguishable. Five angles were used: 0° indicated that the two sea cucumbers were parallel to each other with tentacles or anterior end oriented in the same direction; 45° meant the two sea cucumbers were slightly angled but still oriented in the same direction; 90° was used for sea cucumbers forming a clear right angle with each other; 135° was when the two sea cucumbers were angled with tentacles almost in the opposite direction, and finally, 180° indicated that the two sea cucumbers were parallel in head-to-tail fashion. For each pair analysed, vectors of the closest distance between juveniles were drawn to assess if the orientation of the juveniles was linked to their distance.

2.6. Inter-specific relationships

The same 47 photos used for the analysis of the microhabitat colonised (see section 2.4.) were used to identify the other species

present near the juvenile sea cucumbers on the same stones. The number of juveniles and of the other species were recorded for each photo.

2.7. Statistical analysis

Statistical analyses were conducted in R (R Core Team, 2024). Normality and homoscedasticity were verified using Shapiro-Wilk and Bartlett tests, respectively. Analysis of variance (ANOVA) followed by Tukey HSD post hoc test were used to analyse the effect of depth on the length and density of early recruits; and the influence of size class on the

distance between individuals and their relative orientation. Moreover, a bivariate analysis between the abundance of juveniles and abundance of other species was conducted using the Pearson correlation. Results are presented as mean $\pm$ standard error.

3. Results

3.1. Habitat characterisation

Based on the videos from October 2024, the nursery of Kataaluk was

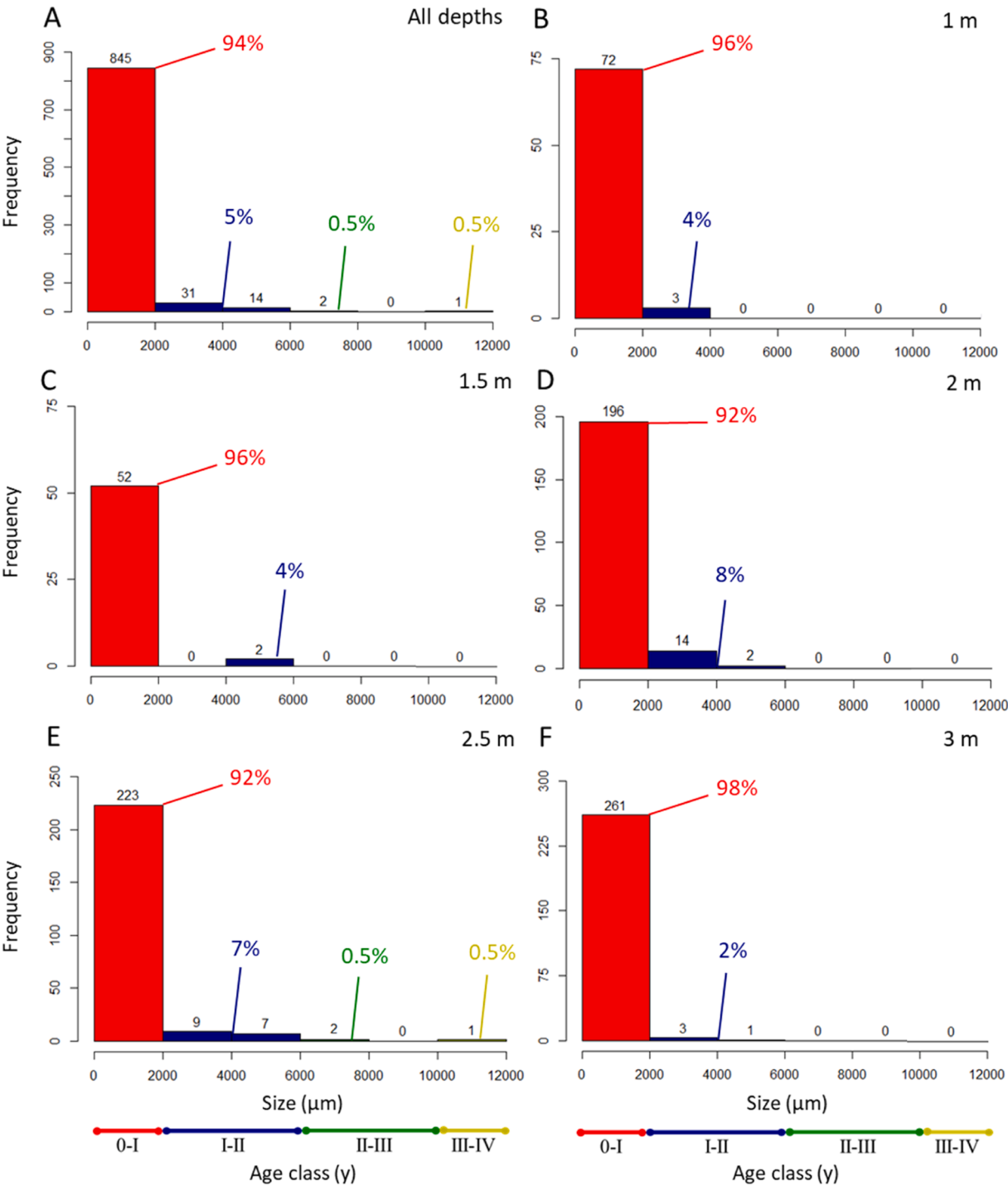


Fig. 2. Population size structure of *Cucumaria frondosa* in Kataaluk (Qikiqitait, Nunavut, Canada) in October 2024 for (A) all depths combined, and depths of (B) 1 m, (C) 1.5 m, (D) 2 m, (E) 2.5 m and (F) 3 m. Estimated age class (in years) of the various size classes as defined by Hamel et al. (2023) are shown. Value above each bar represents the total number of juveniles for each size class.

dominated by rubble, which represented $43.6 \pm 4.6\%$ of all substrata, followed by gravel with $24.1 \pm 2.6\%$, and small boulders with $13.8 \pm 2.4\%$ (Supplementary Figure S2A). Fine sediments were also observed comprising $8.9 \pm 4.8\%$ of the habitat, and large boulders represented $5.0 \pm 2.4\%$ (Supplementary Figure S2A). The least abundant substratum observed was seaweed with $4.7 \pm 5.7\%$ (Supplementary Figure S2A). The summary of the habitat occupied by the juveniles of *C. frondosa* in Kataaluk can be found in Supplementary Figure S3. Environmental parameters were fairly consistent at ~ 0.5 m depth, with seawater temperature ranging between 8.3 and 8.4 °C, salinity between 28.73 and 28.75 psu, and pH between 8.09 and 8.15.

3.2. Population size-structure

There were 4.0 ± 3.6 stones collected per dip netting using the long pole net (minimum: 1; maximum: 14) in October 2024. From them, a total of 78 stones (min length: 1.7 cm; max length: 36 cm) were collected of which 48 ($\sim 62\%$) had sea cucumbers attached to their underside for a total of 893 juveniles of *C. frondosa*.

Measurements of all juveniles from all depths revealed that 845 out of the 893 (95%) had a size ≤ 2000 μm (Fig. 2A) and were therefore classified as early recruits between 0 and 1 year of age. Their size was 1499 ± 9 μm (range: 697–2000 μm contracted length). Most (84%) were found between 2 and 3 m depths; specifically, 24% at 2 m, 28% at 2.5 m and 32% at 3 m. At shallower depths, 10% occurred at 1 m and 6% at 1.5 m (Fig. 2). Additionally, 45 larger juveniles measuring between > 2000 and 6000 μm in length (3063 ± 187 μm) were found at 2 m (39%) and 2.5 m (39%); with smaller proportions observed at 1 m

(7%), 1.5 m (5%) and 3 m (10%; Fig. 2). Finally, two juveniles measuring 6075 and 6828 μm , and a single individual measuring 10,512 μm , were found at 2.5 m depth (Fig. 2E).

When comparing both sampling dates, 95% of the juveniles collected were early recruits in October 2024, with this percentage decreasing to 52% in May 2025 (Fig. 3A and B). About 5% of the juveniles had a size between > 2000 and 6000 μm in October 2024, while almost half of them (46%) were that size in May 2025. Fewer juveniles with a size between > 6000 and 10,000 μm were found in October 2024 (0.4%) than in May 2025 (3%). The mean size of the juveniles increased from 1598 ± 21 μm in October 2024 to 2406 ± 73 μm in May 2025, representing a global increment of $> 50\%$ (~ 800 μm) in seven months, or a growth rate of ~ 114 $\mu\text{m mo}^{-1}$.

The size distribution of the early recruits in October 2024 was normal with a peak between 1400 and 1600 μm in length containing 35% of the individuals (Fig. 3C). The size of the early recruits followed a sigmoid distribution in May 2025 and showed 80% of the early recruits measuring > 1400 μm (Fig. 3D).

3.3. Vertical distribution of early recruits

The length of early recruits was 1558 ± 24 μm at 1 m depth, 1576 ± 30 μm at 1.5 m and 1602 ± 16 μm at 2 m. At 2.5 m, the length was 1483 ± 18 μm , and 1409 ± 15 μm at 3 m (Fig. 4A). They were significantly smaller at 3 m depth (ANOVA, $F_{4,798} = 19.52$, $p < 0.001$; Tukey-HSD, $p < 0.01$). The smallest (697 μm) and longest (2000 μm) individuals were found at 3 m (Fig. 4A).

The density of early recruits at all depths combined was estimated at

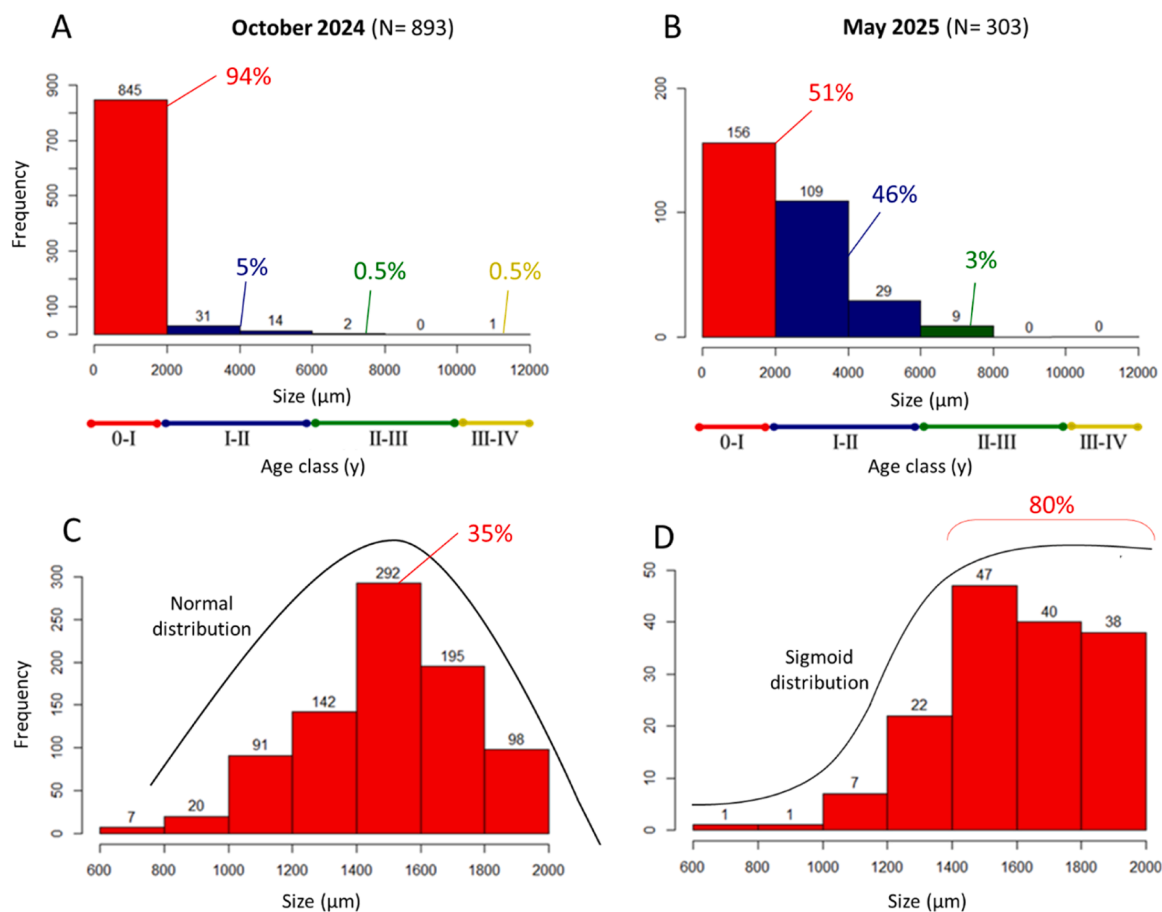


Fig. 3. Population size structure of *Cucumaria frondosa* in Kataaluk (Qikiqtaik, Nunavut, Canada) in (A) October 2024 and (B) May 2025. Estimated age class (in years) of the various size classes as defined by Hamel et al. (2023) are shown. Size structure of the early recruits of *C. frondosa* in (C) October 2024 and (D) May 2025. Value above each bar represents the total number of juveniles for each size class.

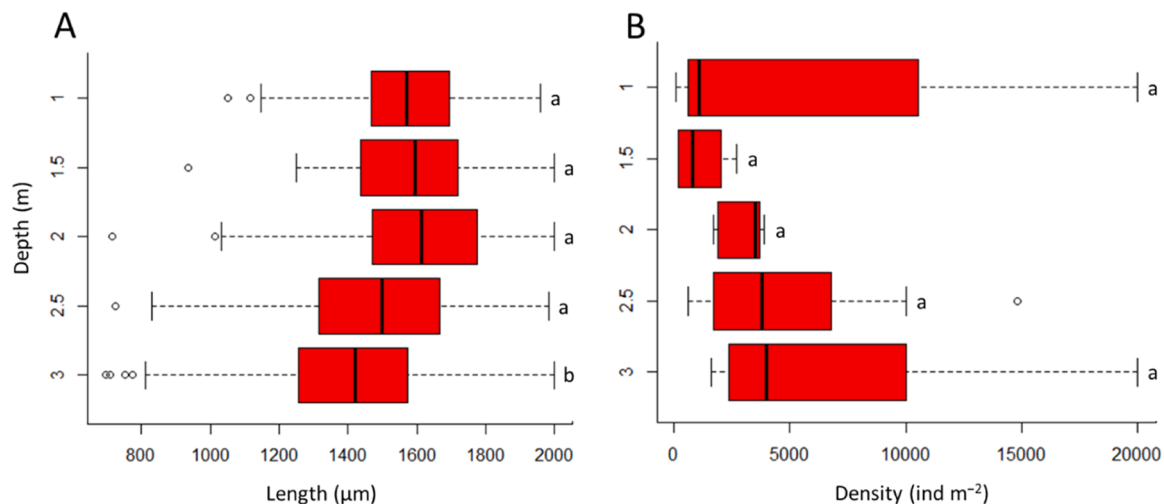


Fig. 4. (A) Length (μm) and (B) density (ind m^{-2}) of the early recruits of *Cucumaria frondosa* according to the sampling depth (m) at Kataaluk (Qikiqtaik, Nunavut, Canada) in October 2024. Different superscript letters indicate significantly different inter-depth results (ANOVA followed by Tukey HSD, $p < 0.05$). From the left to the right, whiskers represent the minimum and the maximum values, left and right borders of the boxplot represent respectively the lower and upper quartiles, the black vertical line represents the median and the open circles represent the outliers.

$7400 \pm 1,000 \text{ ind m}^{-2}$ (minimum: 100; maximum: 50,000 ind m^{-2}). Density was maximal at 2 m with $11,000 \pm 8000 \text{ ind m}^{-2}$ and lower at 1.5 m with $5000 \pm 4000 \text{ ind m}^{-2}$ but it fluctuated between 6700 ± 2000 and $7100 \pm 6400 \text{ ind m}^{-2}$ at 1, 2.5 and 3 m depths (Fig. 4B). Overall, no significant statistical differences were detected among depths (ANOVA, $F_{4,46} = 0.26$, $p > 0.05$).

3.4. Distribution on the substrata and micro-habitat occupancy

A total of 43 out of 48 stones with juveniles attached to their surface were measured (length). One stone on which juveniles were found was classified as a small boulder (36 cm in diameter), and was the only stone with seaweeds on its exposed surface (Supplementary Figure S2B). On this particular boulder, a total of 72 juveniles, including 69 early recruits $\leq 2000 \mu\text{m}$ and three larger individuals measuring between > 2000 and $6000 \mu\text{m}$ were identified (Supplementary Figure S2C). Additionally, 21 stones were classified as gravel (size: $4.8 \pm 0.4 \text{ cm}$, Supplementary Figure S2B), on which 95 individuals were observed, including 89 early recruits and six larger juveniles (Supplementary Figure S2C). Finally, 21 stones were classified as rubble (size: $14.5 \pm 0.9 \text{ cm}$, Supplementary Figure S2B), on which 673 individuals were found, including 640 early recruits and 32 larger juveniles ($> 2000 \mu\text{m}$), including the largest, which was $10,512 \mu\text{m}$ long (Supplementary Figure S2C).

A study of 656 juveniles out of 847 collected in October 2024 (representing 77% of the population studied; all depths combined) showed variability in the structural features of the substrata they occupied (Fig. 5). Of them, 97% were found on the undersides of the stones, in different micro-habitats: 411 individuals (63%) were found on flat surface (Fig. 5A), 121 individuals (18%) in shallow depressions (Fig. 5E), 59 (9%) along the edge of stones (Fig. 5C), and 47 (7%) inside crevices (Fig. 5D). The other 18 juveniles (3%) were observed on the top of the stone but hidden inside the holdfast of seaweeds (Fig. 5B).

3.5. Inter-individual distances and orientations

On a surface area of 1000 cm^2 analysed across ten stones (ten grids of $10 \times 10 \text{ cm}^2$, Supplementary Figure S1), 258 juveniles were found within a surface of 177 cm^2 , indicating that 823 cm^2 of the surface examined had no juveniles. The index of dispersion of nine of the ten stones analysed was superior to 1 (average $I = \sim 1.62$), indicating that juveniles tended to be aggregated, with aggregation sizes fluctuating between 10 juveniles on a surface of 6 cm^2 and 40 juveniles on a surface

of 22 cm^2 (Table 1). The index of dispersion was 0.96 for the remaining stone, where 27 juveniles were randomly distributed over a surface of 24 cm^2 (Table 1).

When considering solely the nearest neighbour, a distance of $0.37 \pm 0.02 \text{ cm}$ was found. The nearest neighbour distance between early recruits ($0.37 \pm 0.01 \text{ cm}$) was higher than that between early recruits and individuals between > 2000 and $6000 \mu\text{m}$ ($0.32 \pm 0.06 \text{ cm}$). Moreover, 43 distances ($\sim 12\%$ of the 350 measurements for the nearest neighbour) were equal to zero, which means that juveniles were touching each other. A total of 34 distances between early recruits and nine distances between early recruits and juveniles from larger size-classes were equal to zero. Groups of up to 10 early recruits touching each other were observed (Fig. 6C). Also, two juveniles, with size ranging between > 2000 and $6000 \mu\text{m}$, were found touching by their posterior ends.

A total of 2884 distances between juveniles were measured based on 17 stones analysed (see examples in Supplementary Figure S4). Overall, the mean distance between a sea cucumber and all other individuals found on a same stone was $3.12 \pm 0.04 \text{ cm}$ (all size classes combined). The distance separating early recruits on a same stone was $3.15 \pm 0.05 \text{ cm}$ ($n = 2702$). The distance between early recruits and individuals > 2000 – $6000 \mu\text{m}$ on a same stone was $2.69 \pm 0.16 \text{ cm}$ ($n = 147$), and the distance between early recruits and the longest individuals $> 6000 \mu\text{m}$ was $2.58 \pm 0.19 \text{ cm}$ ($n = 33$) (Fig. 6A). The distances between the different size-classes of juveniles were similar (ANOVA, $F_{2,2884} = 3.66$, $p > 0.05$). Finally, the distribution of the distances between early recruits were unimodal and right-skewed (Fig. 6B).

In all 150 pairs of early recruits analysed to assess the orientation of their whole body, 68 pairs (134 ind) were oriented in the same direction forming angles between 0 and 45° with each other. Thirty-seven pairs (72 ind) were found in an opposite direction of each other forming angles between 135 and 180° (Fig. 7A). Moreover, 90 individuals (45 pairs) were found perpendicular to each other (angles formed by their anterior ends were 90° ; Fig. 7A). The distance between early recruits facing the same direction (i.e. angle of 0°) was $0.22 \pm 0.07 \text{ cm}$, and it was $0.14 \pm 0.03 \text{ cm}$ for individuals facing opposite directions (i.e. angle of 180° ; Fig. 7B). For the acute, perpendicular and obtuse orientations, the distances between early recruits were respectively 0.20 ± 0.02 , 0.20 ± 0.02 and $0.30 \pm 0.15 \text{ cm}$ (Fig. 7B). The distance between early recruits had no significant influence on their orientation (different angles formed by their anterior ends; ANOVA, $F_{4,148} = 1.08$, $p > 0.05$).

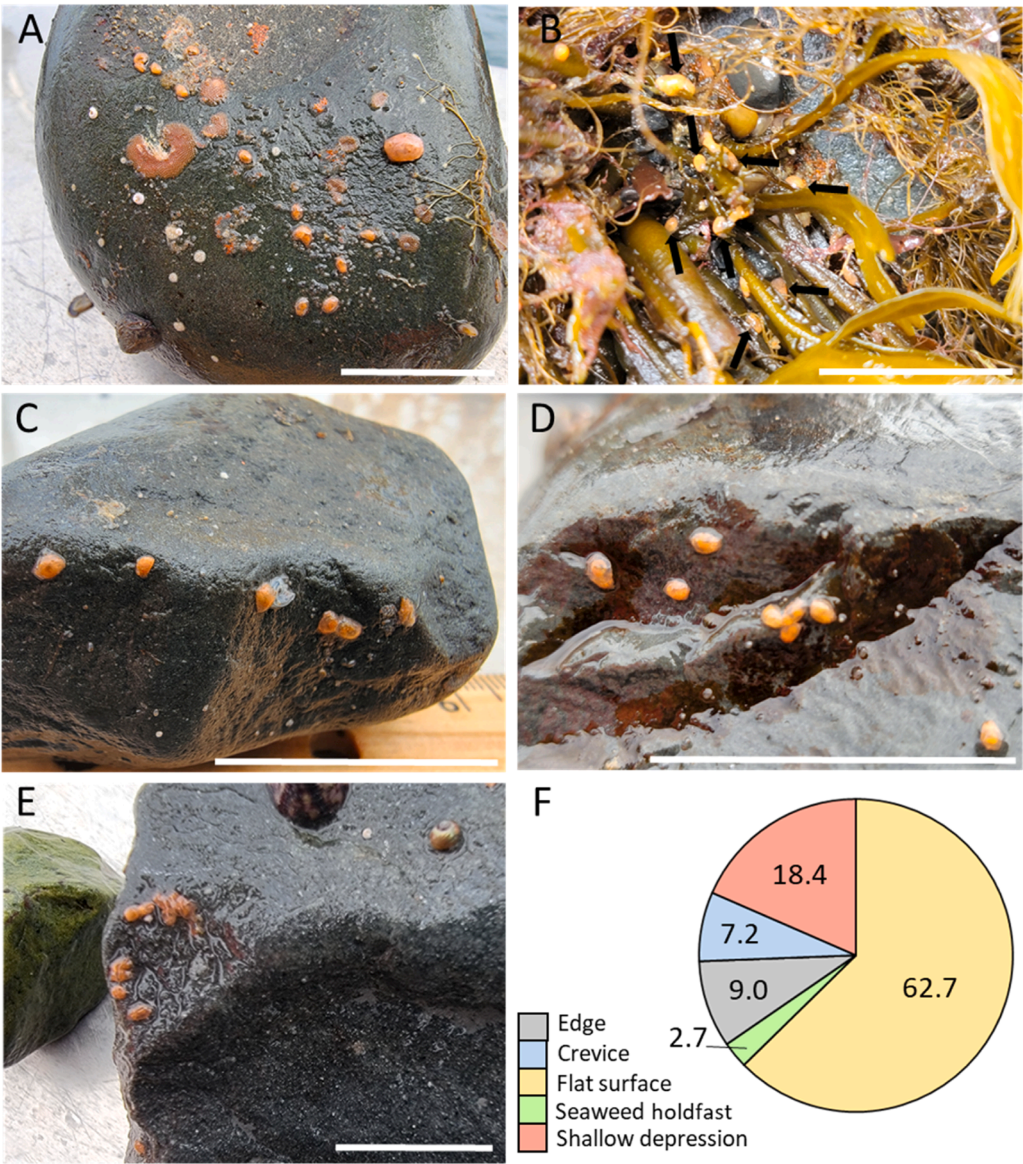


Fig. 5. Different micro-habitats occupied by early recruits of *Cucumaria frondosa* collected at Kataaluk (Qikiqtait, Nunavut, Canada) in October 2024. (A) Individuals on a flat surface of the underside of a stone; (B) on the holdfast of the seaweed (black arrows); (C) lined along a stone edge; (D) in a crevice; (E) in a shallow depression. Scale bars = 2 cm. (F) Repartition of micro-habitat occupancy (%) by juveniles of *Cucumaria frondosa*.

Table 1
Results of the variance-to-mean ratio analyses and associated type of dispersion of the juveniles of *Cucumaria frondosa* on the stones of Kataaluk (Qikiqtait, Nunavut, Canada).

Stone	Mean	Variance (S^2)	Index of dispersion (I)	Type of dispersion	Colonised surface (cm^2)	Number of juveniles
01	0.23	0.60	2.62	Aggregated	12	28
02	0.29	0.43	1.48	Aggregated	22	40
03	0.11	0.18	1.63	Aggregated	8	12
04	0.36	0.43	1.21	Aggregated	27	36
05	0.33	0.36	1.11	Aggregated	27	34
06	0.27	0.26	0.96	Random	24	27
07	0.09	0.16	1.82	Aggregated	6	10
08	0.23	0.34	1.48	Aggregated	16	23
09	0.13	0.20	1.50	Aggregated	10	13
10	0.35	0.59	1.70	Aggregated	25	35

3.6. Species co-occurring on the same stone

Among 656 juveniles, 95.1% (624 ind) were observed with co-occurring species on the underside of the same stone, while only 4.9%

(32 ind) were observed without any other species. A positive linear correlation ($r = 0.63$; $p < 0.001$) was highlighted between the abundance of juvenile sea cucumbers and the abundance of other species found on the same substratum.

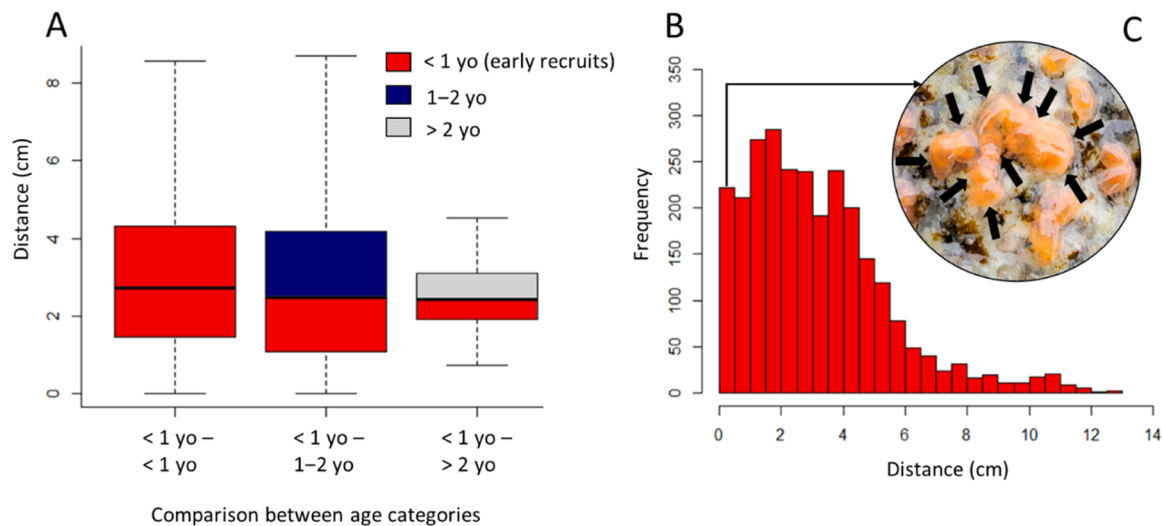


Fig. 6. (A) Distance (cm) between juveniles of *Cucumaria frondosa* from different age-classes found on the stones collected in Kataaluk (Qikiqtait, Nunavut, Canada) in October 2024. Yo: year-old. Whiskers represent the minimum and the maximum values, bottom and top of the boxplot represent respectively the lower and upper quartiles, the black horizontal line represents the median. (B) Frequencies of the distances (cm) between the early recruits of *Cucumaria frondosa*. (C) Close-up of a group of 10 early recruits ($\leq 2000 \mu\text{m}$; arrows) found touching each other (distance = 0 cm) on stones at Kataaluk (Qikiqtait, Nunavut, Canada).

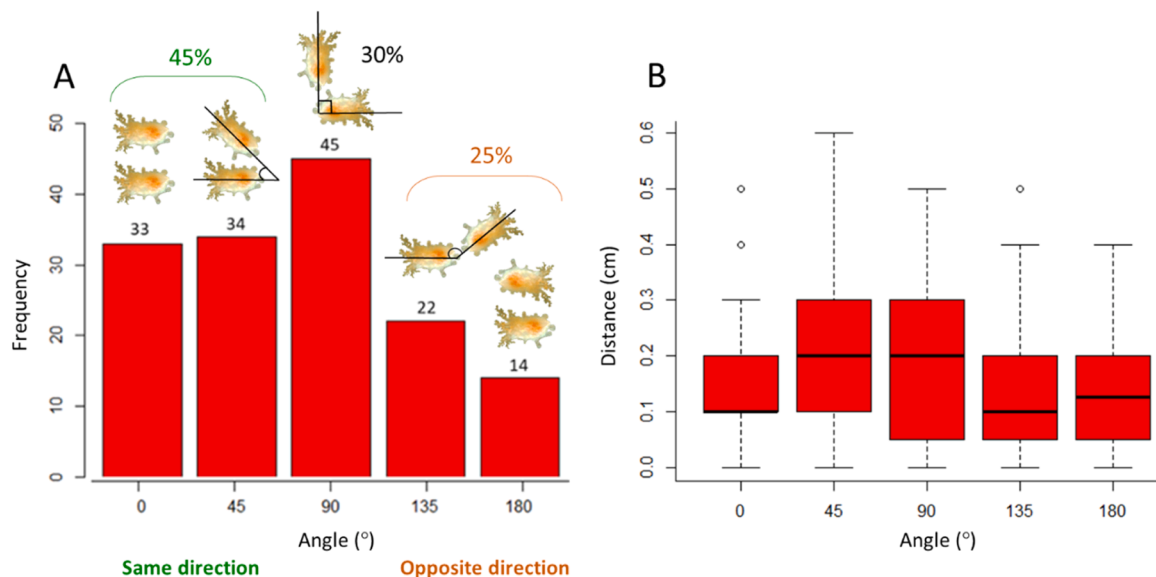


Fig. 7. Relative positioning of early recruits of *Cucumaria frondosa* on stones in Kataaluk (Qikiqtait, Nunavut, Canada) in October 2024. (A) Frequency of each category of angle (°) formed by the anterior part of two recruits; values above each bar represent the total number of pairs of juveniles for each angle. (B) Distance (cm) between early recruits according to the angle (°) formed by their anterior parts. Whiskers represent the minimum and the maximum values, bottom and top of the boxplot represent respectively the lower and upper quartiles, the black horizontal line represents the median and the open circles represent the outliers.

Four taxa co-occurred with the juvenile sea cucumbers under the same stone: the polychaete tubeworm *Spirorbis spirorbis* (Fig. 8A), encrusting bryozoans in the order Cheilostomatida (Fig. 8B), the limpet *Testudinalia testudinalis* and calcareous encrusting algae (Fig. 8C). The abundance of these species on the underside of stones was 758 ind, including 22 encrusting algae (2.9%), 52 limpets (6.9%), 307 bryozoans (40.5%) and 377 tube worms (49.7%; Fig. 8D). In ~96% of the co-occurring observations, the juvenile sea cucumbers and the other taxa identified were close to each other ($< 1 \text{ cm}$), while in the other ~4%, juveniles were touching either the tubeworm or the encrusting bryozoan (Fig. 8B).

4. Discussion

4.1. Spawning season, growth rate and age estimations

The spawning season of *C. frondosa* in Qikiqtait has not been determined precisely yet. In the present study, the size of the juveniles was ~1500 μm (contracted preserved size for the October 2024 collection) and based on the pelagic larval duration for this lecithotrophic species (~40–45 days; Hamel and Mercier, 1996a), it is estimated that spawning occurred around the end of June. Since spawning coincides with the full moon (Mercier and Hamel, 2010) during the phytoplankton bloom (Hamel and Mercier, 1996c), which is around June–July in Hudson Bay (Ferland, Gosselin and Starr, 2011; Matthes et al., 2021), it is proposed that spawning in the spring of 2024 occurred between 18 and 23 June.

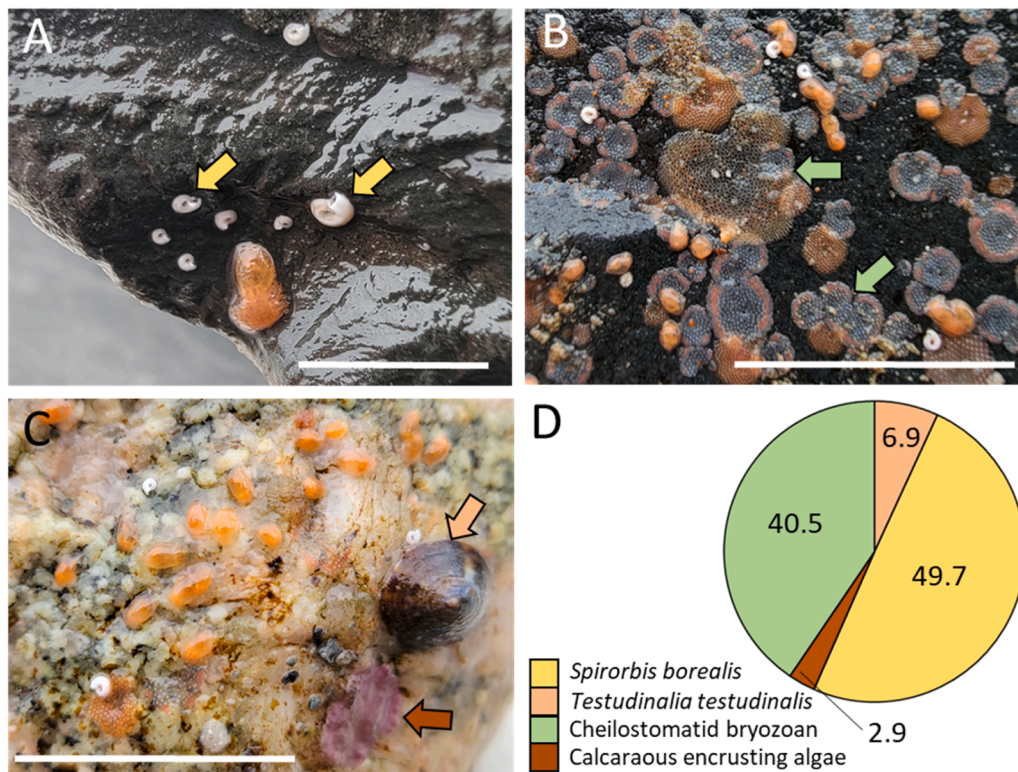


Fig. 8. Sympatric taxa found on the same stones as juveniles of *Cucumaria frondosa* in Kataaluk (Qikiqtait, Nunavut, Canada) in October 2024, including (A) several individuals of the polychaete tubeworm *Spirorbis spirorbis* (yellow arrows), (B) colonies of encrusting bryozoans (green arrows) in the order Cheilostomatida and (C) a limpet *Testudinalia testudinalis* (orange arrow) and patch of calcareous encrusting algae (brown arrow). Scale bars = 2 cm. (D) Diversity (%) of the species found co-occurring on stones with juveniles of *C. frondosa* in Kataaluk.

This places the event a few weeks earlier than proposed in a preliminary study in the same region (Hamel et al., 2023), also roughly aligning with estimates of June–July for the eastern coast of Iceland (Christophersen, Jonasson and Sunde, 2024). Based on this date, the early recruits collected in October 2024 in Qikiqtait would be ~15 weeks post fertilisation (including the six weeks spent in the plankton). Moreover, few small juveniles < 1500 μm were also collected in May 2025. This could indicate that the spawning season of *C. frondosa* in the Canadian Arctic, instead of being a massive single event as in the St. Lawrence Estuary (Québec; Hamel and Mercier, 1996b, 1996c), might extend over several weeks in the summer, as already suggested for populations in Newfoundland, Canada (Coady, 1973; Mercier and Hamel, 2010), in the Barents Sea, Russia (Dvoretzky and Dvoretzky, 2021), and in the Gulf of Maine, USA (Medeiros-Bergen et al., 1995).

Both collections of juveniles (October and May) at the same location (Kataaluk) and depth likely correspond to the same cohort examined seven months apart, suggesting a growth rate of $\sim 114 \mu\text{m mo}^{-1}$ in Qikiqtait (Canadian Arctic). When converted to live size (30–35% longer), this value is similar to or slightly lower than those recorded for the same species held under flow-through mesocosm conditions, i.e. $160 \mu\text{m mo}^{-1}$ by So et al. (2010) and $\sim 230 \mu\text{m mo}^{-1}$ by Gianasi et al. (2018) over the first two years of life. Differences in seasonal temperature cycles (Arctic vs. cold-temperate climate), phytoplankton quality and abundance (principal food source for *C. frondosa*; Hamel and Mercier, 1998) and other physico-chemical factors (field vs. lab) likely play a role in the variation observed. Based on this growth rate and the mean size of 15-week-old juveniles collected in October 2024, it is expected that one-year-old juveniles reached a size of $\sim 2750 \mu\text{m}$ (contracted length), and that juveniles collected in May 2025 ($\sim 2400 \mu\text{m}$) were ~ 10 months old. Factoring in the preserved state, this size is again comparable to or slightly smaller than that of laboratory-reared sea cucumbers after 12 months, i.e. $\sim 3000 \mu\text{m}$ in So et al. (2010) and $\sim 3600 \mu\text{m}$ in

Gianasi, Hamel and Mercier (2018).

The sexual maturity of *C. frondosa* in the Canadian Arctic has been estimated to occur when individuals reach $\sim 130 \text{ mm}$ after ~ 6 years (Hamel et al., 2023). However, the size at sexual maturity was established at ~ 3 – 4 years or a size of 80 – 110 mm based on previous laboratory and field monitoring in eastern Canada (Hamel and Mercier, 1996a). The growth rate recorded in the field for *C. frondosa* in eastern Canada, i.e. $1500 \mu\text{m mo}^{-1}$ (based on the five first years of growth, suggesting that the growth of the sea cucumber follows an exponential curve), and the contracted length ($2750 \mu\text{m}$) of one-year-old juveniles in Qikiqtait, suggest that 4–5 years is necessary to reach the size at sexual maturity. This time is longer by ~ 1 year than that established in the St. Lawrence Estuary (Quebec, eastern Canada), likely due to a difference in primary productivity (principal food source of *C. frondosa*) and environmental factors that could favour higher growth and development rates at lower latitudes.

4.2. Evidence of ontogenetic migration

Distinct elements from the present study support an ontogenetic migration of *C. frondosa* from first settlement depth (determined to be within 3 m) to deeper locations. Despite the smallest size individuals ($\leq 2000 \mu\text{m}$ in length) were the only size found at 1 and 1.5 m , sea cucumbers between > 2000 and $6000 \mu\text{m}$ in length were found at 2 – 5 m and individuals $> 6000 \mu\text{m}$ strictly at 2.5 m . Moreover, only 1% of the individuals collected in October 2024 were larger than $6000 \mu\text{m}$ and only 3% in May 2025. No individual $> 10,500 \mu\text{m}$ was collected in October 2024 and May 2025 nor observed on ROV footage (unpublished data and Ma et al., 2025a). This would suggest that individuals of *C. frondosa* undergo a first migration to exit the nursery upon reaching a size of $\sim 6000 \mu\text{m}$, as they start to develop brown pigmentation on their body wall, as described by Gianasi et al. (2018), which is regarded as

protection against UV radiation. The present study therefore provides evidence of ontogenetic migration from the early stages inside the nursery of Qikiqtait, while it was already demonstrated for larger size classes in the same region. Ma et al. (2025a) showed that individuals between 6 and 11 cm long dominated at depths between 3 and 11 m, while Morrison et al. (2025) showed the presence of solely larger individuals (10–17 cm) in deeper waters down to 23–40 m. The only remaining unknown is the timing of the transition between the cryptic nursery stage, where juveniles up to ~6000 µm are found, and those ≥ 10,000 µm visible on the exposed surface of the substratum (Ma et al., 2025a). Finding populations of individuals measuring between 6000 and 10,000 µm would be the next logical step.

4.3. Intra- and inter-specific relations inside the nursery

The study suggests gregariousness during settlement and early life since 90% of the juveniles were found close to each other (mean nearest neighbour distance: ~0.37 mm), including about 1.5% that were touching each other, forming groups of up to ten juveniles. It was hypothesised by Doll et al. (2022) that in echinoderms, an individual already present on the substratum may cause others to settle close by, because they are either attracted by similar environmental cues or by the presence of conspecifics (Morgan and Jangoux, 2004). Similar gregarious behaviour of early recruits was found in other sea cucumbers (Young and Chia, 1982), brittle stars (Morgan and Jangoux, 2004), sea urchins (Scheibling and Robinson, 2008), as well as in barnacles (Head et al., 2003) and tubeworms (Toonen and Pawlik, 2001). In the sea cucumber *Holothuria scabra*, Claereboudt et al. (2023) demonstrated that sediments and waters where conspecific adults were found attracted juveniles.

Juveniles of *C. frondosa* were found in Kataaluk inside various microhabitats, but always underneath the stones, including on flat surfaces, in shallow depressions or crevices, and along edges. The preference for undersides was also evidenced under laboratory conditions by Hamel and Mercier (1996a). Here, the flat surfaces without asperities were favoured by 63% of the early recruits, which differs from the laboratory results of Hamel and Mercier (1996a) where the percentage of definitive settlement was higher on rocky substratum with crevices (between 38% and 46%) than on smooth surfaces (20–22%). These differences could be attributed to the fact that most stones available for settlement in Kataaluk had flat surfaces without broken areas, crests or crevices, suggesting that larvae and early recruits are relatively flexible in the choice of substrata but might prefer crevices when available, although the contribution of hydrodynamic conditions cannot be ruled out.

No clear pattern was found for the relative body orientation of the juveniles, with 45% of them facing the same direction, 30% facing perpendicular directions and 25% facing opposite directions. Accordingly, juveniles were aggregated but randomly positioned relative to each other. As suspension feeders, individuals of *C. frondosa* passively intercept food particles from the plankton (Hamel and Mercier, 1998) and individuals, despite being sometimes close to each other, need space to optimally extend their tentacles. This, added to the complexity of the three-dimensional substratum composed of piled stones, which likely makes for spatially unpredictable currents and flows of planktonic food, may explain the random positioning of the juveniles.

In total, 95% of the juvenile sea cucumbers were found close to another species, while 4% of them were touching allospecifics. The two most abundant taxa identified to cohabit on the underside of stones were a tubeworm (~50%), encrusting bryozoans (~40%), and a limpet (~7%). The polychaete and bryozoans are active filter feeders (Dales, 1957; Riisgård, Manríquez, 1997; Larsen, Matlok and Riisgård, 1998; Riisgård and Larsen, 2010) and the limpet is a grazer (Arrontes et al., 2004; Coleman et al., 2006). While the different taxa that form the benthic community under the stones in the nursery of Kataaluk possess different feeding strategies, those relying on suspended particles tended to congregate, perhaps identifying areas where planktonic food sources

were most abundant.

4.4. Comparison of the nursery habitats in Kataaluk and Kataapik

Despite their close geographic proximity, the identified nurseries along the coasts of Qikiqtait are not homogenous based on their bathymetric limits, the density and size of juveniles they host, and the diversity of substrata available for them to colonise. Juveniles of *C. frondosa* ≤ 2000 µm in length composed the majority of juveniles in Kataaluk between 1 and 3 m depths. However, the study by Hamel et al. (2023) in Kataapik, which is less than 10 km west of Kataaluk, reported mostly juveniles between 2000 and 6000 µm in length at depths from 1.8 to 2.5 m. Two non-exclusive reasons might explain this difference: 1) the abundance of ducks (mostly eiders) (Jamieson, Robertson and Gilchrist, 2001) known to prey on mussels (Merkel et al., 2006; Rintala et al., 2022), among which most early recruits were recorded in Kataapik (Hamel et al., 2023), and 2) Kataapik being a preferred area for Inuit to harvest mussels and other marine invertebrates for consumption year long, especially in winter. In contrast, the duck and human disturbances are absent in Kataaluk.

When all depths are combined, the density of juveniles in Kataaluk was ~7400 ind m⁻² compared to ~18–23 ind m⁻² previously determined in Kataapik (Hamel et al., 2023), representing much denser population of juveniles at the site studied here. Kataapik is a shallow and narrow channel characterised by strong currents with marked salinity and temperature variations, mainly during the spring thaw, where juvenile sea cucumbers have been found mostly attached to the byssus or shells of live or dead blue mussels as well as on stones of various sizes (Hamel et al., 2023). In Kataaluk, the studied area was outside the wider channel of Tasiujaq Bay where the bottom is strictly covered by stones of different sizes, on which all juveniles were observed. Blue mussels were absent from that site. The difference in density in favour of the stone substratum that dominated at Kataaluk could suggest that this habitat is more suitable for larval settlement and establishment of high densities of early recruits of *C. frondosa*. This is supported by early laboratory work (Hamel and Mercier, 1996a) demonstrating that the pentactula larvae favoured stone and gravel for settlement over all other substrata tested. Moreover, the substratum characterising Kataaluk is in fact composed of more than one layer of stones creating a three-dimensional maze of channels, nooks and crevices that could offer ideal refuge to early recruits while allowing water circulation facilitating the flow of oxygenated water and food to the juveniles (particulate organic matter and phytoplankton; Hamel and Mercier, 1998). Dense populations of green sea urchins (*Strongylocentrotus droebachiensis*), which are known to undertake destructive grazing (Hagen, 1995; Lauzon-Guay and Scheibling, 2007), have also been observed in Kataapik (unpublished ROV footage), while only few of them were observed in Kataaluk (unpublished ROV footage). Gianasi et al. (2021) synthesised the various predators of *C. frondosa* and mentioned that early recruits were susceptible to sea urchin grazing, which could also explain the much lower density of juveniles in Kataapik.

4.5. The extent of the nursery and potential abundance of juveniles

Hamel et al. (2023) estimated that ~4000 km of the 5000 km coastline of Qikiqtait was suitable habitat for early recruits of *C. frondosa* by identifying mussel beds looking at satellite photos of the coastline; they established the colonisable surface area for the early life stages to be between ~16,000 and 24,000 km². Accordingly, they extrapolated that the number of juveniles could be ~300 billion across the entire archipelago. This number was likely underestimating the number of juveniles along the coasts of Qikiqtait, since it was based on a lower juvenile density than recorded here. Moreover, the depth range colonised by the early recruits was extended in the present study (0.8 m shallower and 0.5 m deeper). Integrating the higher density, the wider depth distribution found here, and the new habitat (piled stones), the

number of juveniles in Qikiqtait is likely higher than previously predicted. The present study highlights that the nurseries of *C. frondosa* across Qikiqtait could be heterogeneous. With these two studies, ~2 km of coastline has been explored, leaving most of the coast unexplored. Therefore, to have a more accurate estimation of the number of juveniles around Qikiqtait and to have a better understanding of the variability of their density across the archipelago, additional sampling at different locations are required. Exploring more possible nursery areas and describing them will hopefully help set the boundaries of future marine protected area(s) in this region while supporting subsistence food for the Inuit community. Increasing knowledge on the demographics of early stages is essential when developing a small-scale commercial fishery, in this case led by the Inuit community of Sanikiluaq, considering that the carrying capacity of the fishery depends mainly on the survival of the early stages and their successful migration into the harvested populations.

CRedit authorship contribution statement

Joséphine Pierrat: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Jean-François Hamel:** Writing – review & editing, Methodology, Conceptualization. **Ma Kevin C. K.:** Writing – review & editing, Data curation, Conceptualization. **Annie Mercier:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.rsma.2026.104920](https://doi.org/10.1016/j.rsma.2026.104920).

Data availability

Data will be made available on request.

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